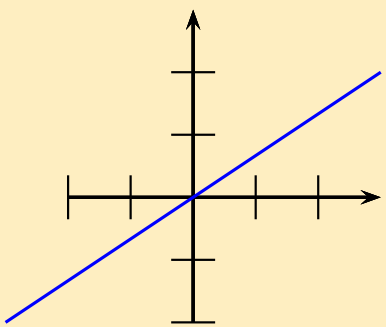
Subspaces of $\mathbb{R}^n$

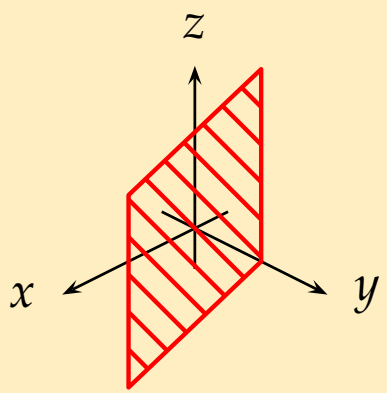
A subspace of $\mathbb{R}^n$ is a set H of vectors in $\mathbb{R}^n$ such that

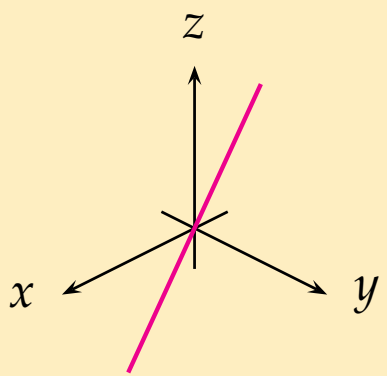
1. The zero vector "$\mathbf{0}$" is in H.
2. For every two vectors $\mathbf{u}$ and $\mathbf{v}$ in H, the sum $\mathbf{u} + \mathbf{v}$ is also in H.
3. For each vector $\mathbf{u}$ in H and every scalar c, the vector $c\mathbf{u}$ is also in H.

Examples of subspaces

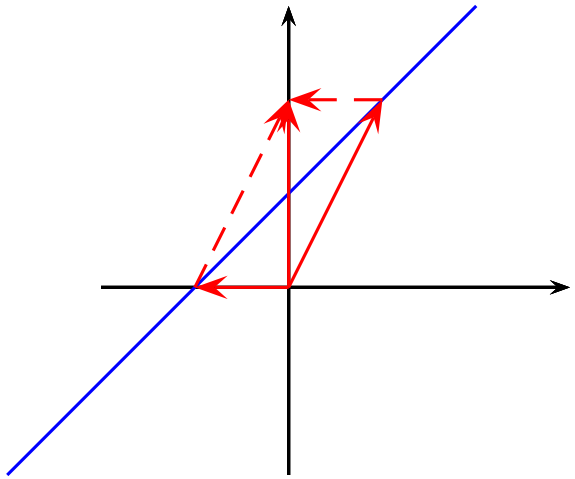
- If $\mathbf{a}_1, .., \mathbf{a}_m$ are vectors in $\mathbb{R}^n$ then Span$\{\mathbf{a}_1, ..., \mathbf{a}_m\}$ is a subspace of $\mathbb{R}^n$.
  - Every line in $\mathbb{R}^n$ that passes through the origin is a subspace of $\mathbb{R}^n$.
  - Every plane in $\mathbb{R}^n$ that passes through the origin is a subspace of $\mathbb{R}^n$.







However, a line or a plane that does not pass through the origin is not a subspace.



A subspace of $\mathbb{R}^n$ is a set H of vectors in $\mathbb{R}^n$ such that

1. The zero vector "$\mathbf{0}$" is in H.
2. For every two vectors $\mathbf{u}$ and $\mathbf{v}$ in H, the sum $\mathbf{u} + \mathbf{v}$ is also in H.
3. For each vector $\mathbf{u}$ in H and every scalar c, the vector $c\mathbf{u}$ is also in H.

Example

Let $H = \left\{ \begin{bmatrix} x \\ y \\ 0 \end{bmatrix} \;\middle|\; x, y \text{ are in } \mathbb{R} \right\}$. Then H is a subspace of $\mathbb{R}^3$.

1. $x = y = 0 \Rightarrow \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ is in H.
2. $\mathbf{u} = \begin{bmatrix} x \\ y \\ 0 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} x' \\ y' \\ 0 \end{bmatrix} \Rightarrow \mathbf{u} + \mathbf{v} = \begin{bmatrix} x \\ y \\ 0 \end{bmatrix} + \begin{bmatrix} x' \\ y' \\ 0 \end{bmatrix} = \begin{bmatrix} x + x' \\ y + y' \\ 0 \end{bmatrix}$ is in H.
3. $\mathbf{u} = \begin{bmatrix} x \\ y \\ 0 \end{bmatrix} \Rightarrow c\mathbf{u} = \begin{bmatrix} cx \\ cy \\ 0 \end{bmatrix}$ is in H.

Basis of a subspace H of $\mathbb{R}^n$

Suppose H is a subspace of $\mathbb{R}^n$. A list of vectors $\mathbf{a}_1, ..., \mathbf{a}_k$ in $\mathbb{R}^n$ is called a basis of H if:

1. $\mathbf{a}_1, ..., \mathbf{a}_k$ are in H.
2. $\text{Span}\{\mathbf{a}_1, ..., \mathbf{a}_k\} = H$.
3. $\mathbf{a}_1, ..., \mathbf{a}_k$ are linearly independent.

Example. The vectors $\begin{bmatrix}1\\0\\0\end{bmatrix}$, $\begin{bmatrix}0\\1\\0\end{bmatrix}$, and $\begin{bmatrix}0\\0\\1\end{bmatrix}$ form a basis of $\mathbb{R}^3$.

(It is called the standard basis of $\mathbb{R}^3$.)

- Every vector in $\mathbb{R}^3$ is a linear combination of them :

$$\begin{bmatrix}x\\y\\z\end{bmatrix} = x\begin{bmatrix}1\\0\\0\end{bmatrix} + y\begin{bmatrix}0\\1\\0\end{bmatrix} + z\begin{bmatrix}0\\0\\1\end{bmatrix}.$$

- Easy to check that $\begin{bmatrix}1\\0\\0\end{bmatrix}$, $\begin{bmatrix}0\\1\\0\end{bmatrix}$, and $\begin{bmatrix}0\\0\\1\end{bmatrix}$ are linearly independent.

Basis of a subspace H of $\mathbb{R}^n$

Suppose H is a subspace of $\mathbb{R}^n$. A list of vectors $\mathbf{a}_1, ..., \mathbf{a}_k$ in $\mathbb{R}^n$ is called a basis of H if:

1. $\mathbf{a}_1, ..., \mathbf{a}_k$ are in H.
2. $\text{Span}\{\mathbf{a}_1, ..., \mathbf{a}_k\} = H$.
3. $\mathbf{a}_1, ..., \mathbf{a}_k$ are linearly independent.

Example

Let $\mathbf{a} = \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix}$ and $H = \text{Span}\{\ \mathbf{a}\ \}$. Then $\{\ \mathbf{a}\ \}$ is a basis of H.

1. $\mathbf{a}$ is in H.
2. $H = \text{Span}\{\ \mathbf{a}\ \}$
3. The set $\{\ \mathbf{a}\ \}$ is linearly independent.

- In fact if $\mathbf{b}$ is any nonzero vector in $\mathbb{R}^n$ then $\{\mathbf{b}\}$ is a basis for $\text{Span}\{\mathbf{b}\}$.

Example

Let $H = \left\{ \begin{bmatrix} x \\ y \\ 0 \end{bmatrix} \;\middle|\; x, y \text{ are in } \mathbb{R} \right\}$. Then H is a subspace of $\mathbb{R}^3$. (Earlier today.)

- The list of vectors $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\}$ is a basis of H :
  - Every vector $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ 0 \end{bmatrix}$ in H is in $\text{Span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\}$:

$$\left[\begin{array}{cc|c} 1 & 1 & b_1 \\ 0 & 1 & b_2 \\ 0 & 0 & 0 \end{array}\right]$$

    Because the rightmost column can never become a pivot column.

  - The vectors $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ are linearly independent.

    Because every column of the matrix $\begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$ has a pivot.

The column space of a matrix : Col A

Suppose A is an $m \times n$ matrix. We can write A as

$$A = [\ \mathbf{a}_1\ \mathbf{a}_2\ \cdots\ \mathbf{a}_n\ ]$$

$\operatorname{Col} A$ is the subspace of $\mathbb{R}^m$ spanned by the columns of A.

$$\operatorname{Col} A = \operatorname{Span}\{\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n\}$$

- $\operatorname{Col} A$ is the set of all vectors $\mathbf{b}$ in $\mathbb{R}^m$ for which the matrix equation $A\mathbf{x} = \mathbf{b}$ is consistent.

Question.

How can we find a basis for Col A?

Algorithm for finding a basis of Col A

Suppose A is an $m \times n$ matrix.

1. Find the R.E.F. (or just the E.F.) of A.
2. The pivot columns of A (not the E.F.) form a basis of Col A.

Example. Find a basis for Col A where $A = \begin{bmatrix} -1 & 2 & 1 & 1 \\ 2 & -4 & 0 & -4 \\ -1 & 2 & 3 & -1 \end{bmatrix}$

First we row reduce A :

$$\begin{bmatrix} -1 & 2 & 1 & 1 \\ 2 & -4 & 0 & -4 \\ -1 & 2 & 3 & -1 \end{bmatrix} \xrightarrow{\text{Row Reduction}} \begin{bmatrix} 1 & -2 & 0 & -2 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The pivot columns are the first and the third.

Therefore a basis for Col A is given by $\mathbf{a}_1 = \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix}$ and $\mathbf{a}_3 = \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}$

i.e., $\left\{ \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} \right\}$ is a basis of Col A.

Example (cont.)

Question. Why does this work?

$$\begin{bmatrix} -1 & 2 & 1 & 1 \\ 2 & -4 & 0 & -4 \\ -1 & 2 & 3 & -1 \end{bmatrix} \xrightarrow{\text{Row Reduction}} \begin{bmatrix} 1 & -2 & 0 & -2 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

We can write the columns $\mathbf{a}_2$ and $\mathbf{a}_4$ as a linear combination of the basis $\{\mathbf{a}_1, \mathbf{a}_3\}$.

Example

$\mathbf{a}_2$: In R.E.F. the second column is $\begin{bmatrix} -2 \\ 0 \\ 0 \end{bmatrix}$ and

$$\begin{bmatrix} -2 \\ 0 \\ 0 \end{bmatrix} = -2 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 0 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

This means that $\mathbf{a}_2 = -2\mathbf{a}_1 + 0\mathbf{a}_3$, i.e.

$$\begin{bmatrix} 2 \\ -4 \\ 2 \end{bmatrix} = -2 \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix} + 0 \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}$$

Example (cont.)

Question. Why does this work?

$$\begin{bmatrix} -1 & 2 & 1 & 1 \\ 2 & -4 & 0 & -4 \\ -1 & 2 & 3 & -1 \end{bmatrix} \xrightarrow{\text{Row Reduction}} \begin{bmatrix} 1 & -2 & 0 & -2 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

We can write the columns $\mathbf{a}_2$ and $\mathbf{a}_4$ as a linear combination of the basis $\{\mathbf{a}_1, \mathbf{a}_3\}$.

Example

$\mathbf{a}_4$: In R.E.F. the fourth column is $\begin{bmatrix} -2 \\ -1 \\ 0 \end{bmatrix}$ and

$$\begin{bmatrix} -2 \\ -1 \\ 0 \end{bmatrix} = (-2)\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + (-1)\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

This means that $\mathbf{a}_4 = (-2)\mathbf{a}_1 + (-1)\mathbf{a}_3$, i.e.

$$\begin{bmatrix} 1 \\ -4 \\ -1 \end{bmatrix} = (-2)\begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix} + (-1)\begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}$$

Justification

- Row operations preserve linear relations between columns.
- For example, consider a sequence of elementary row operations :

$$[\ \mathbf{a}_1\ \mathbf{a}_2\ \cdots\ \mathbf{a}_n\ ] \xrightarrow{\text{Some elementary row operations}} [\ \mathbf{b}_1\ \mathbf{b}_2\ \cdots\ \mathbf{b}_n\ ]$$

  - If $\mathbf{a}_2 - 3\mathbf{a}_3 + 2\mathbf{a}_5 - \mathbf{a}_6 = 0$ then $\mathbf{b}_2 - 3\mathbf{b}_3 + 2\mathbf{b}_5 - \mathbf{b}_6 = 0$
  - If $\mathbf{b}_3 = -2\mathbf{b}_1 + 3\mathbf{b}_4$ then $\mathbf{a}_3 = -2\mathbf{a}_1 + 3\mathbf{a}_4$.
  - If $\mathbf{a}_1, \mathbf{a}_3, \mathbf{a}_6$ are linearly independent, then $\mathbf{b}_1, \mathbf{b}_3, \mathbf{b}_6$ are also linearly independent (and vice versa).

As a result we have :

Theorem

The pivot columns of any matrix A form a basis for $\operatorname{Col} A$.

Example

Find a basis for $\operatorname{Col} A$ where $A = \begin{bmatrix} 3 & -2 & 4 & 0 \\ 9 & -6 & 12 & 0 \\ 6 & -4 & 8 & 1 \end{bmatrix}$.

Solution.

$$\begin{bmatrix} 3 & -2 & 4 & 0 \\ 9 & -6 & 12 & 0 \\ 6 & -4 & 8 & 1 \end{bmatrix} \xrightarrow{\text{Row Reduction}} \begin{bmatrix} 1 & -\frac{2}{3} & \frac{4}{3} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The pivot columns are the first and the fourth.

Therefore a basis for $\operatorname{Col} A$ is:

$$\left\{ \begin{bmatrix} 3 \\ 9 \\ 6 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Example

Suppose $A = \begin{bmatrix} 1 & 0 & 0 & 2 & -3 & 0 \\ 0 & 0 & 1 & -5 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$. Find a basis for $\operatorname{Col} A$.

Solution. A is in R.E.F. and the pivot columns are the first, third and sixth columns.

It follows that $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$ is a basis of $\operatorname{Col} A$.

Example

Let $A = \begin{bmatrix} 1 & -5 & 1 \\ -2 & 12 & 4 \\ -3 & 14 & -6 \end{bmatrix}$.

Find a basis for the subspace H of $\mathbb{R}^3$ containing all vectors $\mathbf{b} \in \mathbb{R}^3$ such that the matrix equation $A\mathbf{x} = \mathbf{b}$ is consistent.

$$H = \{\mathbf{b} \in \mathbb{R}^3 \ : \ A\mathbf{x} = \mathbf{b} \text{ is consistent}\}$$

Solution. We know that

$$\operatorname{Col} A = \{\mathbf{b} \in \mathbb{R}^3 \ : \ A\mathbf{x} = \mathbf{b} \text{ is consistent}\} = H.$$

Therefore finding a basis for H is equivalent to finding a basis for $\operatorname{Col} A$.

$$\begin{bmatrix} 1 & -5 & 1 \\ -2 & 12 & 4 \\ -3 & 14 & -6 \end{bmatrix} \xrightarrow{\text{Row reduction}} \begin{bmatrix} 1 & 0 & 16 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

Therefore a basis for $H = \operatorname{Col} A$ is

$$\left\{ \begin{bmatrix} 1 \\ -2 \\ -3 \end{bmatrix}, \begin{bmatrix} -5 \\ 12 \\ 14 \end{bmatrix} \right\}$$

Basis for Nul A

Nul A

Suppose A is an $m \times n$ matrix.

The null space of A is the subspace of $\mathbb{R}^n$ containing all $\mathbf{x}$ such that $A\mathbf{x} = \mathbf{0}$.

The null space of A is denoted by Nul A.

Algorithm for finding a basis of Nul A

1. Solve the linear system $A\mathbf{x} = \mathbf{0}$ in the usual way, i.e., by finding R.E.F. of $[\ A \mid 0\ ]$.
2. Write the solution in vector parametric form :

$$\mathbf{x} = x_{i_1}\mathbf{c}_1 + \cdots + x_{i_r}\mathbf{c}_r$$

3. The vectors $\mathbf{c}_1, \ldots, \mathbf{c}_r$ will be a basis of Nul A.

Example

Find a basis for $\operatorname{Nul} A$ where $A = \begin{bmatrix} 1 & -1 & 0 & -2 & 0 \\ 2 & -2 & 1 & -1 & 0 \\ 1 & -1 & -1 & -5 & 1 \end{bmatrix}$

Solution. To solve the system $A\mathbf{x} = \mathbf{0}$ we need to find the R.E.F. of the following augmented matrix :

$$\left[\begin{array}{ccccc|c} 1 & -1 & 0 & -2 & 0 & 0 \\ 2 & -2 & 1 & -1 & 0 & 0 \\ 1 & -1 & -1 & -5 & 1 & 0 \end{array}\right] \xrightarrow{\text{Row Reduction}} \left[\begin{array}{ccccc|c} 1 & -1 & 0 & -2 & 0 & 0 \\ 0 & 0 & 1 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{array}\right]$$

x_1, x_3, x_5 : basic

x_2, x_4 : free

$$\begin{cases} x_1 = x_2 + 2x_4 \\ x_3 = -3x_4 \\ x_5 = 0 \end{cases} \implies \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} x_2 + 2x_4 \\ x_2 \\ -3x_4 \\ x_4 \\ 0 \end{bmatrix} = x_2 \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 2 \\ 0 \\ -3 \\ 1 \\ 0 \end{bmatrix}$$

And a basis for $\operatorname{Nul} A$ is given by $\begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ -3 \\ 1 \\ 0 \end{bmatrix}$

Example

Find a basis for Nul A where $A = \begin{bmatrix} -1 & 2 & 1 & 1 \\ 2 & -4 & 0 & -4 \\ -1 & 2 & 3 & -1 \end{bmatrix}$.

Solution. We solve the system $A\mathbf{x} = \mathbf{0}$:

$$\left[\begin{array}{cccc|c} -1 & 2 & 1 & 1 & 0 \\ 2 & -4 & 0 & -4 & 0 \\ -1 & 2 & 3 & -1 & 0 \end{array}\right] \xrightarrow{\text{Row Reduction}} \left[\begin{array}{cccc|c} 1 & -2 & 0 & -2 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

x_1, x_3 : basic

x_2, x_4: free

$$\begin{cases} x_1 = 2x_2 + 2x_4 \\ x_3 = x_4 \end{cases}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2x_2 + 2x_4 \\ x_2 \\ x_4 \\ x_4 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 2 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

Basis for Nul A: $\left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 1 \\ 1 \end{bmatrix} \right\}$

Algorithm for finding a basis of Nul A

1. Solve the linear system $A\mathbf{x} = \mathbf{0}$ in the usual way, i.e., by finding R.E.F. of $[\ A \mid 0\ ]$.
2. Write the solution in vector parametric form :

$$\mathbf{x} = x_{i_1}\mathbf{c}_1 + \cdots + x_{i_r}\mathbf{c}_r$$

3. The vectors $\mathbf{c}_1, ..., \mathbf{c}_r$ will be a basis of Nul A.

Why does the algorithm work?

- We know that Nul A = Span$\{\mathbf{c}_1, ..., \mathbf{c}_r\}$.
- $\mathbf{c}_1, ..., \mathbf{c}_r$ are linearly independent : each $\mathbf{c}_i$ has an entry which is equal to 1, but the corresponding entry in the other $\mathbf{c}_j$'s is equal to 0.

$$x_{i_1}\mathbf{c}_1 + \cdots + x_{i_r}\mathbf{c}_r = \begin{bmatrix} \vdots \\ x_{i_1} \\ \vdots \\ x_{i_r} \\ \vdots \end{bmatrix} \qquad x_{i_1}\mathbf{c}_1 + \cdots + x_{i_r}\mathbf{c}_r = \mathbf{0} \implies \begin{bmatrix} \vdots \\ x_{i_1} \\ \vdots \\ x_{i_r} \\ \vdots \end{bmatrix} = 0$$

$$\implies \quad x_{i_1} = \cdots = x_{i_r} = 0$$

Dimension

Dimension of a subspace H

Suppose H is a subspace of $\mathbb{R}^n$.

Fact. All of the bases of H have the same number of vectors.

The number of vectors in a basis of H is called the dimension of H.

Remark. By convention, the subspace $\{\mathbf{0}\}$ has dimension zero.

Example

Let $A = \begin{bmatrix} -1 & 2 & 1 & 1 \\ 2 & -4 & 0 & -4 \\ -1 & 2 & 3 & -1 \end{bmatrix}$. What is the dimension of $\operatorname{Col} A$?

Earlier today we found a basis for $\operatorname{Col} A$: $\mathbf{a}_1 = \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix}$ and $\mathbf{a}_3 = \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}$

It follows that the dimension of $\operatorname{Col} A$ is 2.

Example

What is the dimension of $H = \text{Span}\left\{ \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 6 \\ 0 \\ 3 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix} \right\}$?

Solution. First we find a basis for H. But this is the same as finding a basis for $\text{Col}\, A$ where $A = \begin{bmatrix} 2 & 6 & -2 & 0 & 4 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 3 & -1 & 1 & 2 \end{bmatrix}$

We apply the algorithm : we should find the R.E.F. of A.

$$\begin{bmatrix} 2 & 6 & -2 & 0 & 4 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 3 & -1 & 1 & 2 \end{bmatrix} \xrightarrow{\text{Row reduction}} \begin{bmatrix} 1 & 3 & -1 & 0 & 2 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The first and fourth columns are pivot columns. Therefore

$$\left\{ \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$$

is a basis for H and $\dim H = 2$.

Example

Example

Find a basis for the subspace of $\mathbb{R}^3$ given by the equation $x - 2y + 3z = 0$.

Solution. This is a system of one equation in three variables.
In fact it is the null space of the following matrix :

$$A = \begin{bmatrix} 1 & -2 & 3 \end{bmatrix}$$

To solve the problem we should find a basis for Nul A.

x : basic
y, z : free.

$$x = 2y - 3z \quad \Longrightarrow \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2y - 3z \\ y \\ z \end{bmatrix} = y \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}$$

Therefore a basis for the above subspace is $\left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \right\}$

Rank of a matrix

Suppose A is an $m \times n$ matrix. The rank of A is the dimension of $\operatorname{Col} A$.
We denote the rank of A by $\operatorname{rank} A$.

Example

What is the rank of $A = \begin{bmatrix} 1 & -1 & 0 & -2 & 0 \\ 2 & -2 & 1 & -1 & 0 \\ 1 & -1 & -1 & -5 & 1 \end{bmatrix}$?

Solution.
We row reduce the matrix :

$$\begin{bmatrix} 1 & -1 & 0 & -2 & 0 \\ 2 & -2 & 1 & -1 & 0 \\ 1 & -1 & -1 & -5 & 1 \end{bmatrix} \xrightarrow{\text{Row Reduction}} \begin{bmatrix} 1 & -1 & 0 & -2 & 0 \\ 0 & 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

The pivot columns correspond to elements of the basis of $\operatorname{Col} A$.

There are three pivot columns. Therefore $\operatorname{Col} A$ is three dimensional, i.e. $\operatorname{rank} A = 3$.

Example

Question. In the above example, what is the dimension of $\operatorname{Nul} A$?

Solution. The dimension of $\operatorname{Nul} A$ is the number of free variables of $A\mathbf{x} = \mathbf{0}$.

There are two free variables. Therefore $\underline{\dim \operatorname{Nul} A = 2}$.

The Rank Theorem

Suppose A is an $m \times n$ matrix.

- $\operatorname{rank} A = \dim \operatorname{Col} A$= number of pivot columns of A.
- $\dim \operatorname{Nul} A$ = number of free variables in $A\mathbf{x} = \mathbf{0}$
  = number of non-pivot columns of A.

Therefore since A has n columns, we have

$$\operatorname{rank} A + \dim \operatorname{Nul} A = n$$

Remark. Sometimes $\dim \operatorname{Nul} A$ is called the nullity of A.

Example

Let $A = \begin{bmatrix} 1 & -1 & 0 & -2 & 0 \\ 2 & -2 & 1 & -1 & 0 \\ 1 & -1 & -1 & -5 & 1 \end{bmatrix}$. We saw that

Therefore: $\begin{bmatrix} 1 & -1 & 0 & -2 & 0 \\ 2 & -2 & 1 & -1 & 0 \\ 1 & -1 & -1 & -5 & 1 \end{bmatrix} \xrightarrow{\text{Row Reduction}} \begin{bmatrix} 1 & -1 & 0 & -2 & 0 \\ 0 & 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

- $\operatorname{rank} A = 3$
- $\dim \operatorname{Nul} A = 2$

and

$$\operatorname{rank} A + \dim \operatorname{Nul} A = 3 + 2 = 5 = \text{number of columns of } A.$$

In general a subspace has lots of bases. Our methods above provide one basis.

Basis Theorem

Suppose H is a subspace of $\mathbb{R}^n$ and dim $H = p$. Let $\mathbf{v}_1, ..., \mathbf{v}_p$ be p vectors in H.

- If $\mathbf{v}_1, ..., \mathbf{v}_p$ are linearly independent, then $\mathbf{v}_1, ..., \mathbf{v}_p$ is a basis of H.
- If $\text{Span}\{\mathbf{v}_1, ..., \mathbf{v}_p\} = H$, then $\mathbf{v}_1, ..., \mathbf{v}_p$ is a basis of H

Remark. If $\dim H = p$ and $\mathbf{v}_1, ..., \mathbf{v}_q$ are q vectors such that $q > p$ and $H = \text{Span}\{\mathbf{v}_1, ..., \mathbf{v}_q\}$, then it is NOT true that $\{\mathbf{v}_1, ..., \mathbf{v}_q\}$ is a basis of H.

In fact since $q > p$, the set $\{\mathbf{v}_1, ..., \mathbf{v}_q\}$ will be linearly dependent.

The Invertible Matrix Theorem (extension of the theorem in Section 2.3)

Let A be an $n \times n$ matrix. Then the following statements are equivalent to the statement that A is an invertible matrix.

1. The R.E.F. of A is I_n.
2. A has n pivot positions.
3. The equation $A\mathbf{x} = \mathbf{0}$ only has the trivial solution $\mathbf{x} = \mathbf{0}$.
4. The columns of A form a linearly independent set.
5. For every vector $\mathbf{b}$ in $\mathbb{R}^n$, the equation $A\mathbf{x} = \mathbf{b}$ has a unique solution.
6. The columns of A span $\mathbb{R}^n$.
7. A^T is invertible.
8. The columns of A form a basis of $\mathbb{R}^n$.
9. $\dim \mathrm{Col}\, A = n$.
10. $\mathrm{rank}\, A = n$.
11. $\mathrm{Nul}\, A = \{\mathbf{0}\}$.
12. $\dim \mathrm{Nul}\, A = 0$.

Summary of Algorithms

Algorithm for finding a basis of Col A

Suppose A is an $m \times n$ matrix.

1. Find the R.E.F. (or just the E.F.) of A.
2. The pivot columns of A form a basis of Col A.

Algorithm for finding a basis of Nul A

Suppose A is an $m \times n$ matrix.

1. Solve the linear system $A\mathbf{x} = \mathbf{0}$ in the usual way, i.e., by finding R.E.F. of $[\ A \mid 0\ ]$.
2. Write the solution in vector parametric form :

$$\mathbf{x} = x_{i_1}\mathbf{c}_1 + \cdots + x_{i_r}\mathbf{c}_r$$

3. The vectors $\mathbf{c}_1, \ldots, \mathbf{c}_r$ will be a basis of Nul A.